ENTERPRISE TECHNICAL SPECIFICATION

TB FURNITURES

Modern UK Luxury eCommerce & Enterprise Admin CMS

Comprehensive architectural blueprint, database schema dictionary, Entity-Relationship Diagram (ERD), API specifications, Emporio Armani visual typography implementation, and production deployment topology.

CLIENT / SYSTEM NAME TB Furnitures UK Ltd (Storefront & Admin)	PRODUCTION FRAMEWORK Next.js 14 (App Router) + TypeScript
DATABASE INFRASTRUCTURE Neon Serverless PostgreSQL (Isolated Schema: tb_furnitures)	CLOUD HOSTING & EDGE Vercel Global Edge Network + GitHub CI/CD
SECURITY & AUTHENTICATION NextAuth.js (JWT) + BCrypt + UK GDPR Compliant	DELIVERABLE VERSION & DATE Version 1.0.0 — October 2026 Production Release

SYSTEM ARCHITECTURE & TOPOLOGY

OVERVIEW

TB Furnitures is an enterprise-grade digital flagship designed for the British bespoke and handcrafted furniture sector. The platform integrates high-performance customer shopping experiences with an isolated, secure enterprise administration portal.

> **Key Architecture Highlight:** The platform is engineered with a strict decoupled presentation layer. The customer storefront operates with dynamic 4-palette theme engines and luxury typography, while the administrative control panel runs inside an isolated, distraction-free environment with role-based governance and bulk CSV master data operations.

High-Level System Topology

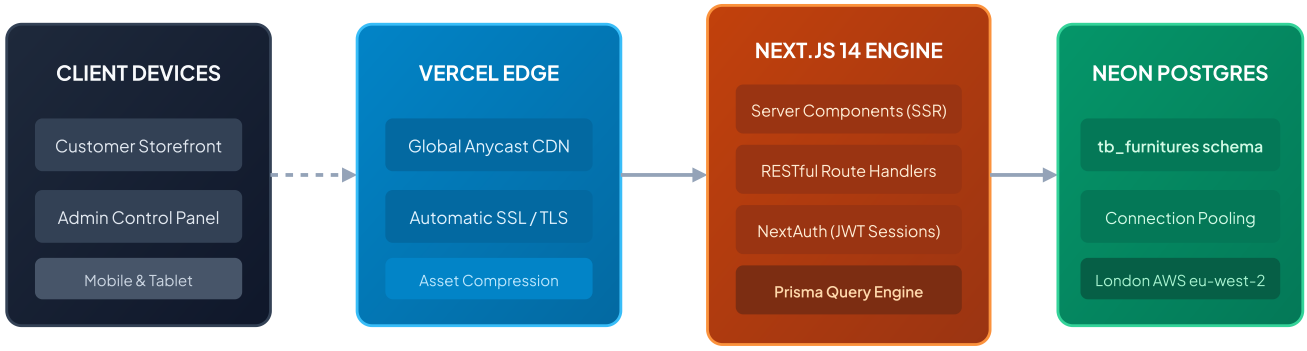


FIGURE 1.0 — END-TO-END ENTERPRISE CLOUD INFRASTRUCTURE & DATA FLOW

Core Technology Stack

LAYER	TECHNOLOGY SELECTED	VERSION	ARCHITECTURAL PURPOSE & ADVANTAGES
Frontend Web	Next.js App Router (React 18)	14.2.35	Streaming SSR, zero client bundle overhead for static views, instant page transitions.
Styling & Design	Tailwind CSS + CSS Custom Properties	3.4.11	Emporio Armani tracked uppercase branding; instant 4-palette dynamic theme switcher.
State & Session	React Context API + NextAuth	4.24.7	Cart persistence across browser sessions, JWT auth tokens, zero external session stores.
Data Access (ORM)	Prisma ORM	5.19.0	Type-safe database client, declarative schema definitions, automated schema migrations.
Database (RDBMS)	Neon Serverless PostgreSQL	16 (Serverless)	Isolated `tb_furnitures` schema namespace, automated pooling, high availability.
Payments (PCI-DSS)	Stripe API SDK	16.10.0	Secure checkout sessions, zero customer card storage on servers, webhook integration.

DATABASE ARCHITECTURE & ENTITY DIAGRAM

SCHEMA

The database schema resides within the dedicated `tb_furnitures` PostgreSQL schema on Neon Serverless. The model guarantees referential integrity, financial precision using exact `Decimal(10,2)` data types, and immutable order snapshots for statutory tax compliance.

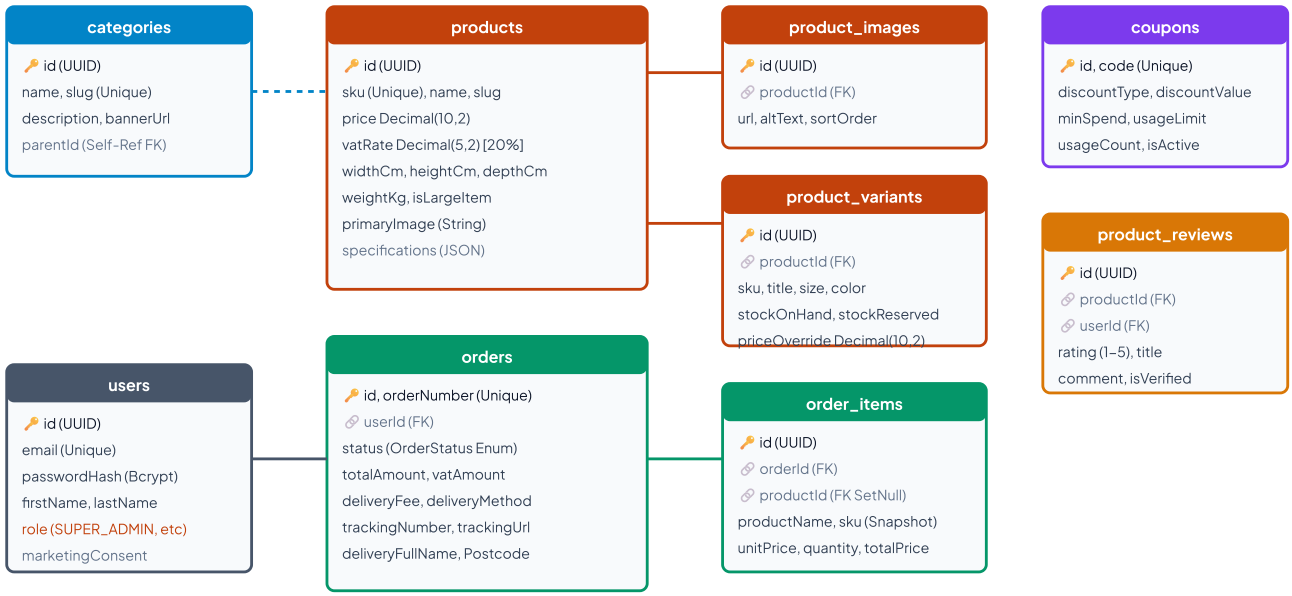


FIGURE 2.0 — ENTITY-RELATIONSHIP DIAGRAM (ERD) WITH FOREIGN KEY CONSTRAINTS

Core Data Dictionary

TABLE NAME	PRIMARY KEY	KEY ATTRIBUTES & CONSTRAINTS	BUSINESS LOGIC & NOTES
`products`	`id` (UUID)	`sku` (Unique), `price` Decimal(10,2), `vatRate` (20.00), `widthCm`, `heightCm`, `depthCm`	Core furniture piece catalogue. Houses physical dimensions for 2-man room delivery calculations.
`product_images`	`id` (UUID)	`productId` (FK), `url` (Text), `sortOrder` (Int)	Supports multi-angle studio galleries (front angle, room setting, and material close-ups).
`categories`	`id` (UUID)	`slug` (Unique), `parentId` (Self-referencing FK)	Hierarchical category taxonomy for furniture rooms (Living Room, Bedroom, Dining Room, etc.).
`orders`	`id` (UUID)	`orderNumber` (e.g. TB-84910), `userId` (FK), `totalAmount`, `status`	Order master with immutable shipping and billing snapshot and courier tracking numbers.
`order_items`	`id` (UUID)	`orderId` (FK), `productId` (FK SetNull), `unitPrice`, `quantity`	Permanent snapshot of item name, SKU, price, and photo at the exact moment of sale.
`users`	`id` (UUID)	`email` (Unique), `passwordHash`, `role` (Enum)	Manages roles: `SUPER_ADMIN`, `ADMIN`, `ORDER_MANAGER`, and `CUSTOMER`.

BACKEND ARCHITECTURE & APIS

API REF

All backend services are implemented as Next.js Route Handlers with type-safe request parsing, JWT authentication guards, and database transactions.

RESTful Endpoints Reference

ENDPOINT ROUTE	METHOD	AUTH GUARD	FUNCTION DESCRIPTION & PAYLOAD
`/api/auth/[...nextauth]`	`POST`	Public	Credentials authentication provider. Validates email and bcrypt password, sets signed HTTP-only JWT.
`/api/products`	`GET`	Public	Fetches products with query params: `?category=`, `?search=`, `?sort=price_asc|desc`.
`/api/products/[slug]`	`GET`	Public	Retrieves product specifications, multi-image gallery, active variants, and verified reviews.
`/api/admin/products`	`GET`	Admin	Full product list with filters, stock levels, and pagination for table rendering.
`/api/admin/products`	`POST`	Admin	Creates or updates furniture items. Also handles bulk CSV import and data transformation.
`/api/admin/categories`	`GET / POST / DELETE`	Admin	CRUD for room categories with automatic slug generation and product relationship validation.
`/api/admin/orders`	`GET`	Admin	Searchable orders table across customer names, postcodes, and order IDs with status filters.
`/api/admin/orders/[id]`	`GET / PUT / DELETE`	Admin	Fetches order details, updates status (e.g. Dispatched + tracking code), or permanently deletes.
`/api/checkout`	`POST`	Customer	Initiates Stripe payment intent or demo order placement with UK VAT and delivery calculation.

Order Fulfillment State Machine



FIGURE 3.0 — ORDER LIFECYCLE & FULFILLMENT STATUS TRANSITION WORKFLOW

FRONTEND ENGINEERING & DESIGN SYSTEM

UI / UX

The customer storefront embodies modern British luxury, taking direct aesthetic inspiration from the geometric uppercase typography and tracking found in high-end design houses (e.g. Emporio Armani).

Emporio Armani Typography Implementation

> **Branding Typography Spec:** Primary brand identifiers, top navigation elements, collection titles, and uppercase banners utilize high-impact geometric grotesque styling (`Plus Jakarta Sans` / `Montserrat`) set at `font-extrabold` with wide letter-spacing `tracking-[0.2em]` or `tracking-[0.25em]` . Body copy is rendered with clean modern sans-serif for reading comfort across high-DPI displays.

4-Palette Dynamic Theme Engine

The user interface features a live theme-switching swatch engine placed directly on the top announcement bar. Themes are applied instantly across the entire application tree without page reloads and persist seamlessly in the user's browser session.

British Heritage (Default)

Warm Cotswold honey, rich chestnut wood tones, and soft amber accents. Best for showcasing handcrafted solid European oak and heritage craftsmanship.

Midnight Noir (High Contrast)

Architectural monochrome styling. Automatically recalculates header typography to pure black or pure white depending on background elevation to ensure strict WCAG AA contrast.

Royal Nordic (British Navy)

Refined maritime navy blue combined with cool slate grey borders and crisp white cards. Ideal for contemporary living rooms and coastal collections.

Modern Sage (Botanical)

Earthy sage green tones with warm stone paper neutrals. Highlights sustainable timber sourcing, organic cotton bouclés, and eco-conscious finishes.

Key Storefront Capabilities

🔍 Live Predictive Search

Real-time dropdown auto-suggestions showing thumbnail, price, and category directly under the search bar, with a dedicated full search results page with sorting and filtering.

🖼️ Multi-Angle Studio Galleries

Interactive multi-image galleries with high-resolution close-ups of wood joinery, fabric weaves, and room context views, paired with specialist 2-man room delivery badges.

🛒 Slide-Over Cart & Checkout

Zero-friction sliding cart drawer with real-time subtotal, UK delivery fee computation (£0 over threshold), and coupon discount redemption.

📦 Customer Account & Tracking

Dedicated customer portal with clickable order numbers, status progression bars, courier tracking hyperlinks, and printable UK VAT receipts.

ADMIN CONTROL PANEL CAPABILITIES

CMS

The Administrative Control Panel is accessible via `/admin` and provides full operational autonomy over master data, inventory levels, order workflows, and customer accounts.

Admin Architecture Highlights

MODULE	KEY FEATURES & ADMIN CAPABILITIES
Dashboard & KPI Analytics	Live summary cards for gross revenue, active orders, total furniture catalog, and registered customer accounts with quick navigation.
Product Catalogue Hub	Comprehensive inventory table with thumbnail previews, SKU lookups, category filters, and quick in-place editing. Supports **CSV Export** of all items and **CSV Import** for bulk uploads.
Order Fulfillment Center	Order search bar, real-time status dropdown update (Dispatched, In Transit, Delivered, Cancelled), courier tracking assignment, and full order detail views with item photo breakdown.
Category Management	Visual hierarchy list with inline Add / Edit / Delete modals. Prevents accidental deletion of categories currently assigned to active products.
Promotions & Coupons	Creation and scheduling of percentage or fixed GBP discounts, minimum spend thresholds, and maximum redemption limits.
User & Role Governance	Administrative team access controls with role assignments (Super Admin, Catalogue Manager, Order Manager) and audit logs.

UK Compliance, Consumer Rights & Tax Regulations

GB Consumer Rights Act 2015

Integrated return policy pages and order receipts fulfilling UK statutory rights for 14-day cooling off periods and 30-day faulty goods remedies.

🛡️ UK GDPR & Data Protection Act 2018

Explicit customer marketing consent checkboxes, secure password hashing (BCrypt 12 rounds), and privacy policy compliance for cookie and analytics tracking.

🧾 HM Revenue & Customs (HMRC) VAT

Transparent 20% Standard UK Value Added Tax calculation with itemized VAT breakdown displayed across checkout and customer invoices.

🚚 Specialist Two-Man Delivery

Automated detection of large/heavy furniture pieces (sofas, extending tables, bed frames) with two-man white-glove delivery tier assignment.

READY FOR PRODUCTION DELIVERY

All code committed to GitHub repository • Neon PostgreSQL database live • Vercel deployment verified